



Getting Started with Python



1. Introduction to Programming

- **Programming** = giving step-by-step instructions to a computer to accomplish a task.
- Instructions are written using a **programming language**.
- The set of instructions is called a **program**.
- Computers follow instructions literally, fast, and precisely — they need exact logic.



2. Why We Learn Programming

- Develops **logical thinking & problem-solving skills** (computational thinking).
- Automate repetitive tasks → save time & effort.
- Gateway to careers in **AI, Data Science, Web Dev, App Dev, Cybersecurity**.
- Builds creativity: games, websites, automation tools, and more.
- Mandatory in CBSE curriculum to nurture future-ready skills.



3. Programming Languages & Classification

- **Low-level languages:** Machine language (0/1), Assembly language – hardware dependent.
- **High-level languages:** Python, Java, C++, JavaScript – human readable, portable.



4. Familiar / Trending Languages (2025 Context)

Language	Primary Use
 Python	AI, ML, Data Science, Web, Automation, Education
 Java	Android apps, Enterprise systems
 C++	Game dev, OS, performance-critical apps
 JavaScript	Web development (frontend + backend)
 Kotlin / Swift	Android / iOS development
 R	Statistics, Data visualization

✓ Python is the most recommended first language due to its simple, English-like syntax and vast community support.

5. Language Processors (Compiler, Interpreter, Assembler)

- **Compiler:** Translates entire source code at once into machine code. Example: C, C++.
- **Interpreter:** Translates and executes code line by line. Python uses an interpreter → easier debugging.
- **Assembler:** Converts assembly language to machine code.

💡 Python's reference implementation **CPython** is written in C and acts as an interpreter + compiler hybrid (bytecode).

6. Basics of Python Programming

◆ Introduction to Python

- High-level, general-purpose, interpreted language.
- Created by **Guido van Rossum** and first released in **February 1991**.
- Named after the BBC comedy show "**Monty Python's Flying Circus**" (1970s).
- CPython is the standard implementation (written in C).

✨ Features of Python

- Easy to read & learn → beginner friendly.
- Interpreted (no separate compilation).
- Cross-platform (Windows, macOS, Linux).
- Open source & free.
- Dynamically typed (no explicit variable type declaration).
- Rich standard library + thousands of third-party modules.
- Supports procedural, object-oriented, and functional programming.

Where is Python used in the real world?

- Web development (Django, Flask)
- Data Science & Analytics (Pandas, NumPy)
- Artificial Intelligence / Machine Learning (TensorFlow, PyTorch)

- Scientific computing, Automation & Scripting
- Cybersecurity, Game development (Pygame)
- Finance, quantitative trading

Applications developed using Python

- **Instagram** (backend)
- **YouTube** (core components)
- **Spotify** (backend & analytics)
- **Netflix** – recommendation system
- **Dropbox** – desktop client
- **Reddit, Uber** (dynamic pricing & algorithms)

7. Execution Modes in Python (Interactive vs Script Mode)

Interactive Mode

- Interacting directly with Python Shell (>>> prompt).
- Instant output after each statement.
- Commands are NOT saved automatically as a file.
- Best for testing small pieces of code and quick experiments.

```
>>> print("Hello Python")
Hello Python
>>> 15 + 30
45
```

Script Mode

- Write multiple lines of code in a **.py** file and execute the entire program.
- Efficient for large programs, reusable code.
- Output appears in console or terminal.

```
# saved as first.py
name = "Class 11"
print("Welcome to Python,", name)
```



8. Executing a Simple "Hello World" Program

In Interactive Mode:

```
>>> print("Hello World")  
Hello World
```

In Script Mode:

- Open any IDE/text editor (IDLE, VS Code, Thonny).
- Type: `print("Hello World")`
- Save as `hello.py` → run using terminal: `python hello.py`
- Output: `Hello World`



9. Where to Write / Run Python Code?

- **Python IDLE** – IDE (built-in).
- **Anaconda Navigator** – Package manager with Jupyter Notebook, Spyder.
- **Jupyter Notebook** – Web-based interactive environment (great for teaching).
- **Spyder** – Scientific IDE.
- **PyCharm** – Professional IDE.
- **Thonny** – Perfect for absolute beginners.
- **VS Code** – Powerful source-code editor with extensions.
- **Sublime Text** – Lightweight editor.



10. Jupyter Notebook – Key Highlights (CBSE friendly)

- Web-based interactive environment, opens in a browser.
- Code organized into **cells** – runs interactively.
- Shortcut to run a cell: `Shift + Enter`.
- Supports **Markdown cells** (text, images, tables) + Code cells (Python).
- Output (charts, tables, images) appears right below the cell.
- Save notebooks as `.ipynb` files, perfect for sharing and reports.
- Great for combining explanation, code, and results in one document.



11. Quick Revision Table

Concept	Key Point
Python Creator	Guido van Rossum
First Release	February 1991
Name Inspiration	Monty Python's Flying Circus (BBC comedy)
Python is ...	Interpreted, high-level, dynamically typed
Two Modes	Interactive Mode & Script Mode
Interactive Mode	Immediate output, not saved
Script Mode	.py file, run whole program
Language Processor for Python	Interpreter (CPython)
Jupyter run cell shortcut	Shift + Enter
Jupyter file extension	.ipynb